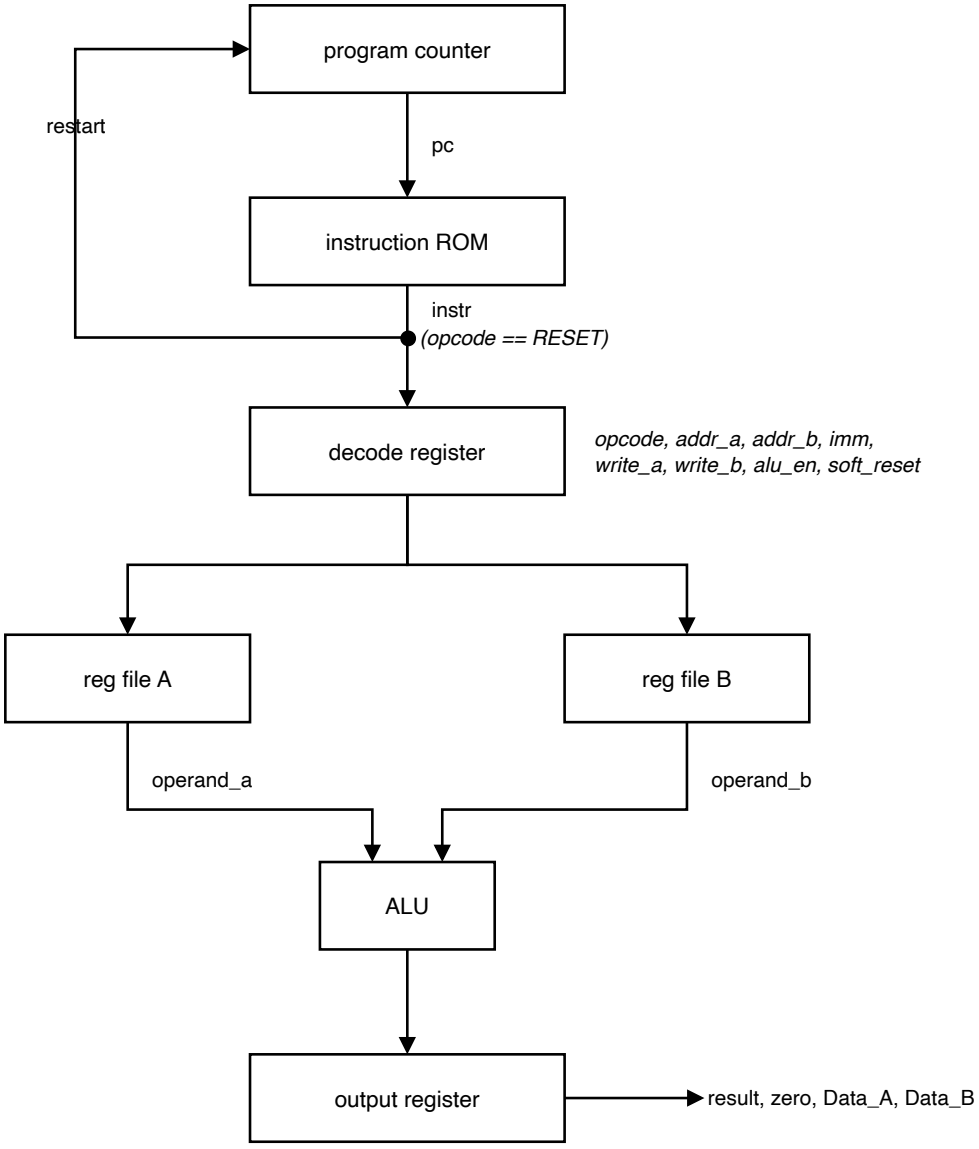
program counter
restart
pc
instruction ROM
instr
(opcode == RESET)
decode register
opcode, addr_a, addr_b, imm,
write_a, write_b, alu_en, soft_reset
reg file A
reg file B
operand_a
operand_b
ALU
output register
result, zero, Data_A, Data_B